

Performance Testing

Date	14 July 2026
Team ID	XXXXXX
Project Name	Smart Lender – AI Based Loan Approval System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	[e.g., JMeter, Locust, k6, Postman]
Type of Testing	[e.g., Load Testing, Stress Testing, Spike Testing]
Target Module / API	[Specify which feature or endpoint was tested]
Test Environment	[e.g., Local / Cloud / Hosted Server]
Test Date	

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Login – Multiple users log in simultaneously	20	60	All users should log in successfully with acceptable response time and no errors.
2	Loan Eligibility Prediction – Multiple users submit loan applications	50	120	The system should process all requests correctly and return loan eligibility results without failures
3	Database Read/Write Operations – Concurrent form submissions	30	90	User data should be stored correctly with no data loss or duplicate records.
4	Peak Load Test – Simultaneous login, loan application, and prediction	100	180	The application should remain responsive, maintain stable performance, and handle concurrent requests without crashing.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.32 seconds	pass	Average response time is within the acceptable limit
2	Response Time (Max)	< 5 seconds	3.84 seconds	pass	Maximum response time remained below the threshold.
3	Throughput (Req/sec)	➤ 50 ➤ Req/sec	68Req/sec	pass	Application handled concurrent requests efficiently.
4	Error Rate	< 1%	0.25%	pass	Very few request failures were observed during testing.
5	CPU Utilization	< 80%	61%	pass	CPU usage remained within acceptable limits
6	Memory Utilization	< 80%	58%	pass	Memory consumption was stable throughout the test.

Step 4: ☐ The Smart Lender application remained stable throughout the performance testing process.

☐ The average response time was well below the target of 2 seconds, providing a smooth user experience.

☐ The maximum response time stayed within the acceptable limit, even under peak load conditions.

☐ The application achieved satisfactory throughput, indicating it could efficiently process multiple user requests simultaneously.

☐ The error rate remained below 1%, demonstrating reliable request handling and minimal failures.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]